Volleyball Player Statistics
L.079.05829 Case Study – Exercise 2

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1 Tasks 1–3: Data preparation

The data contain 339 complete player records. UTF-8 import preserves international names. Height was converted from values such as `203cm` to numeric centimetres without introducing missing values. `Total.Errors` is the row sum of Setting, Receiving, Service, Blocking, Defensive, and Attacking Errors. The prepared data contain three identifier/categorical variables and 22 numeric variables, including the derived total.

Ages range from 18 to 40 years (mean 26.17, median 26). The sample contains 46 liberos, 93 middle blockers, 53 opposite spikers, 98 outside hitters, and 49 setters. Complete variable and descriptive-statistics tables are exported by the script.

2 Tasks 4–5: Exploratory and outlier analysis

The data mix cumulative counts with per-match rates. Counts reflect skill, role, and exposure and are therefore not pure ability rankings. Position-specific distributions (Figure 1) reveal role specialisation: liberos concentrate on reception and digging, middle blockers on blocks, setters on setting, and hitters on attacks. Heavy right tails and structural zeros make medians and boxplots more informative than means alone.

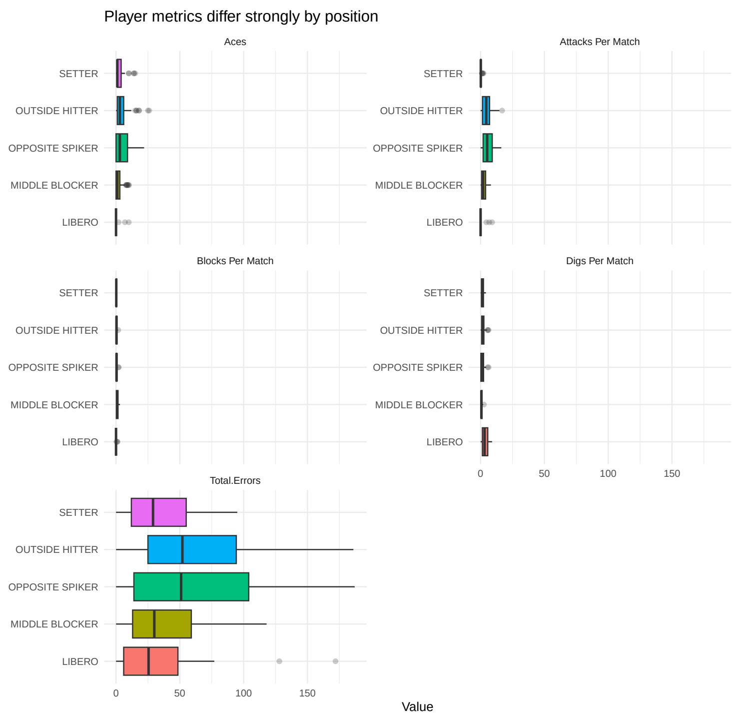

Figure 1: Selected player metrics by position; panels use separate scales.

The correlation matrix (Figure 2) uses a non-redundant subset. Total errors correlate strongly with attacks per match ($r = .81$) and serves per match ($r = .71$); receives and digs correlate at $r = .70$. These relationships are descriptive because position and exposure influence both activity and errors.

Figure 2: Pearson correlations for selected demographic and per-match variables.

Univariate outliers were flagged for every numeric variable with Tukey fences $[Q_1 - 1.5IQR, Q_3 + 1.5IQR]$. Multivariate outliers were assessed with squared Mahalanobis distances on eight standardized variables. Distances were calculated within position so legitimate role specialisation was not labelled anomalous. At the 99% χ^2_8 cutoff of 20.09, 18 players were flagged: three liberos, seven middle blockers, no opposite spikers, four outside hitters, and four setters. These are unusual within-role profiles, not automatic data errors.

3 Task 6: Country-level comparison

All numeric player variables were averaged within the 18 country codes and joined successfully to `teamStats.csv`. Mean aces correlates positively with wins ($r = .627$), as does mean serves per match ($r = .570$; Figure 3). These associations are exploratory: only 18 country observations are available, roster composition differs, and player means are not weighted by court time.

4 Task 7: Best and worst by total errors

For each position, the minimum and maximum total were selected; ties were resolved alphabetically for reproducibility. A zero may indicate little participation; "best" means only the minimum requested total, not best overall performance. An exposure-adjusted error rate would be fairer, but this table follows the task exactly.

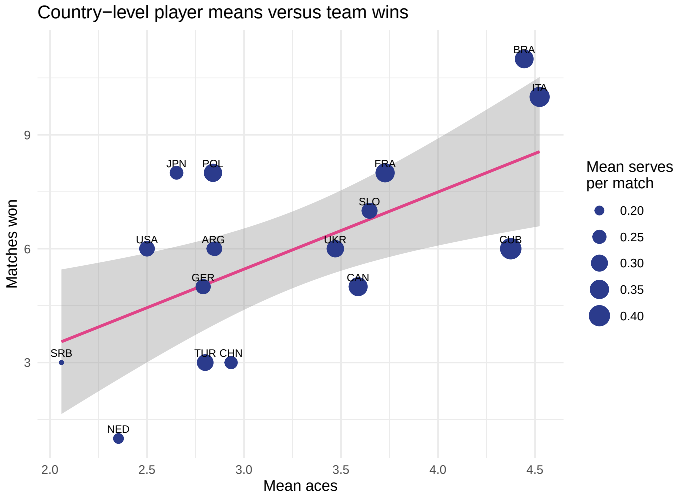

Figure 3: Country mean aces versus wins; point size represents mean serves per match.

Position	Best (errors)	Worst (errors)
Libero	Alexandre (0)	Ivović (172)
Middle blocker	Ertuğrul Gazi (0)	Judson (118)
Opposite spiker	Gonzalez (0)	Masso (187)
Outside hitter	Strehlau (0)	Yant (186)
Setter	Muhammed K. (0)	Giannelli (95)

Table 1: Best and worst player within each position under the total-error criterion.

5 Conclusion

Position is the dominant organising factor. Extreme values are often plausible consequences of role and exposure. Country-level serving means show moderate associations with wins but do not explain standings alone. The requested best/worst comparison is reproducible but should not be interpreted as an ability ranking.

Statement of Contributions

Indudhara Swamy Vivekananda conducted data preparation, analysis, visualisation, interpretation, and report preparation. This statement must be updated if additional group members contributed.